6. Fig. 6.1 shows the variation of binding energy per nucleon for nuclides with a nucleon number greater than 40.

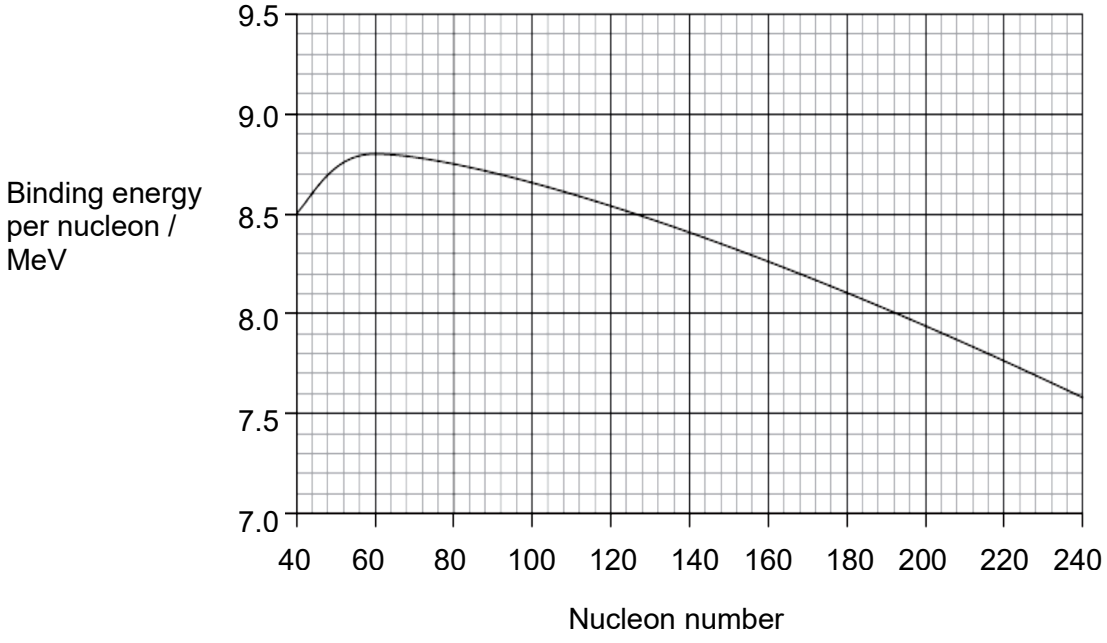

Fig 6.1

The following nuclear reaction occurs when a slow moving neutron is absorbed by an isotope of uranium-235.

$$^{1}_{0}n + ^{235}_{92}U \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + \text{neutrons}$$

(a) Determine the number of neutrons produced in this reaction. [1]

(b) Explain how this reaction is able to produce energy. [2]

(c) Use the graph in **Fig. 6.1** to determine the energy released in the above reaction. [3]

(d) Suggest why the neutron is not included in the calculation above. [1]